Although influenced by the ideas of Malthus, Verhulst realized that there was a major flaw in his predictions. Malthus had claimed that human population tends to increase geometrically, doubling at regular time intervals. Verhulst thought this to be too simplistic, reasoning that the Malthus model did not take into account a larger population's difficulty in finding food. He argued instead that "the population gets closer and closer to a steady state," in which the rate of reproduction is proportionate to both the existing population and the amount of available food. In Verhulst's model, after the point of maximum population growth—the "point of inflection"—the growth rate becomes progressively slower, gradually leveling off to reach the "carrying capacity" of an area—the number of individuals it can sustain. When visualized, Verhulst's model produces an S-shaped curve, which was later called a logistic curve.

"The hypothesis of geometric progression can hold only in very special circumstances."

Pierre-François Verhulst

Practical demonstrations

Verhulst's model was ignored for several decades, partly because he himself was not entirely convinced. However, in 1911, Scottish army physician and epidemiologist Anderson McKendrick used the logistic equation to forecast growth in populations of bacteria. Then, in 1920, Verhulst's equation was adopted and promoted in America by Raymond Pearl.

Pearl conducted experiments with fruit flies and hens. He gave a constant quantity of food to fruit flies kept in a bottle. Initially, their fertility rate increased. However, as the population density grew, competition for resources increased, and eventually reached a bottleneck. After this, the flies' fertility rate dropped; their numbers continued to increase but slowly, and generally the population level stabilized.